ISRO MISSION CONTROL AUTONOMOUS OPERATIONS

SPARC-PM

Comprehensive Software Engineering & System Architecture Submission Report

PROJECT SPECIFICATION & SUBMISSION METADATA

Project Domain: Autonomous Space Operations & Space Weather Survivability
Target Agency: ISRO Telemetry, Tracking and Command Network (ISTRAC) / MCF
Core Capabilities: SGP4 3D Mechanics ? PyTorch LSTM Crew Dosimetry ? CNN-LSTM Failure Prediction ? A* Scheduler
Cryptographic Standard: RFC 2104 HMAC-SHA256 Binary Telecommand Verification
Target System Baseline: Multi-Tier Streamlit Mission Operations Console + Cesium 3D Engine
Submission Version: 2.4.0 (Production Architecture Edition)
Report Classification: Official Engineering Project Submission

EXECUTIVE ABSTRACT

1. PROJECT OVERVIEW & PROBLEM STATEMENT

TABLE OF CONTENTS

1. PROBLEM STATEMENT

1.1 Operational Context & Environmental Hazard Dynamics
The near-Earth orbital environment is governed by dynamic solar-terrestrial coupling. During active solar phases, the Sun emits intense Coronal Mass Ejections (CMEs) and Solar Energetic Particle (SEP) events. The consequences to orbital operations are multi-faceted:
? Relativistic Proton Flux (>10 MeV): High-energy ions penetrate satellite exterior skins, triggering single-event latchups (SEL), permanent CMOS sensor noise, and cumulative ionization damage on gallium-arsenide solar arrays.
? Interplanetary Magnetic Field (IMF) Bz Reconnection: When IMF Bz swings strongly southward (Bz < -10 nT), it reconnects with Earth's dayside geomagnetic field, causing magnetopause standoff compression from ~10 Earth radii down to GEO orbits (35,786 km).
? Thermospheric Atmospheric Drag Spikes: Joule heating expands the upper atmosphere, multiplying aerodynamic drag on Low Earth Orbit (LEO) satellites (Cartosat-3, Oceansat-3, RISAT-2BR1) by factors of 300% to 800%, causing uncontrolled orbital decay.

1.2 The Critical Ground-Station Response Latency Gap
India's Aditya-L1 solar observatory operates at the Sun-Earth Lagrange Point 1 (L1), 1.5 million kilometers upstream of Earth. This strategic vantage point provides a 15-to-45-minute early warning window before coronal shocks reach near-Earth space. However, legacy mission control operations at ground stations face severe structural bottlenecks:
1. Telemetry Silos: Solar payload data (ASPEX, PAPA, MAG) is analyzed independently from satellite housekeeping buses.
2. Manual Flight Surgery Calculation: Radiation absorbed doses for the Gaganyaan crew module are manually integrated over time.
3. High Command Formulation Latency: Evaluating multi-satellite collision corridors and formulating safe-mode telecommands takes 30 to 45 minutes?often arriving after the solar proton shock front has already impacted the fleet.

1.3 Formal Project Objectives & Success Criteria
SPARC-PM directly resolves this operational latency gap through the following quantitative objectives:
? Objective 1: Sub-100ms Autonomous Execution ? End-to-end processing from telemetry ingest to signed command generation.
? Objective 2: 90%+ Radiation Forecasting Accuracy ? 6-hour solar proton flux and Gaganyaan cabin dose rate curve prediction.
? Objective 3: 100% Cryptographic Command Integrity ? Generating RFC 2104 compliant HMAC-SHA256 telecommand frames.

2. USER STORIES & OPERATIONAL PERSONAS

SPARC-PM is built around four primary operational personas stationed at the ISRO Mission Control Center (ISTRAC).

PERSONA 1: DR. VIKRAM SHARMA | GAGANYAAN FLIGHT SURGEON

Domain: Human Spaceflight Crew Health & Radiation Safety
? User Story US-01: As a Flight Surgeon, I want real-time visualization of the Gaganyaan crew cabin radiation dose rate (uSv/hr) and cumulative absorbed dose (mSv), so that I can ensure astronaut safety stays strictly within the 50 uSv/hr safe threshold.
? Acceptance Criteria: Dashboard must display real-time cabin LET integration, career limit progress bars, and automatically trigger an EVA ABORT DIRECTIVE whenever forecasted flux exceeds 250 uSv/hr.
? User Story US-02: As a Flight Surgeon, I want a 6-hour forward-looking radiation forecast curve, so that I can schedule spacewalks (EVA) only during confirmed calm solar intervals.
? Acceptance Criteria: PyTorch LSTM forecast curve rendered with dual-axis overlay (proton flux pfu vs cabin dose rate uSv/hr).

PERSONA 2: ANANYA ROY | ISRO FLEET OPERATIONS LEAD

Domain: Constellation Survivability, 3D Orbital Mechanics & Collision Avoidance
? User Story US-03: As Fleet Operations Lead, I want an interactive 3D Mission Globe displaying active ISRO satellites color-coded by their Euclidean hazard ratio (H_j), so that I can instantly identify which spacecraft are crossing solar storm corridors.
? Acceptance Criteria: Real-time SGP4 orbital propagation of 50 active satellites, Earth surface mesh, and dynamic storm envelope.
? User Story US-04: As Fleet Operations Lead, I want an explainable 'WHY?' rationale for every compromised asset, so that I understand the exact physical telemetry and model outputs driving an alert before approving orbital maneuvers.
? Acceptance Criteria: 3-column breakdown (Model Output, Physics Evidence, Operational Response) available for every satellite.

PERSONA 3: RAJESH KUMAR | SUBSYSTEM RELIABILITY SPECIALIST

Domain: Predictive Maintenance, Degradation Tracking & Component Lifespan
? User Story US-05: As a Subsystem Reliability Specialist, I want early prediction of the 72-hour component failure probability (P_fail) for satellite batteries, gyroscopes, and thermal buses, so that maintenance actions are scheduled prior to catastrophic bus failure.
? Acceptance Criteria: Storm-conditioned 1D CNN-LSTM model computing P_fail between 0.0 and 1.0 across all 50 satellites.
? User Story US-06: As a Subsystem Reliability Specialist, I want a one-click '[+ ADD TO A* QUEUE]' action for any degraded asset, so that corrective telecommands are integrated into the automated scheduling pass.

PERSONA 4: GROUND STATION TELECOMMAND CONTROLLER

Domain: Telecommand Synthesis, Power Budgeting & Uplink Authentication
? User Story US-07: As a Telecommand Controller, I want an autonomous A* action scheduler to prioritize and rank corrective commands without violating onboard power budgets (P_max) or torque limits (T_max), so that the satellite bus is not overloaded.
? Acceptance Criteria: A* priority search executes in < 25 ms, generating an ordered dispatch list bounded by power limits.
? User Story US-08: As a Telecommand Controller, I want every generated telecommand frame cryptographically signed with HMAC-SHA256, so that ground station transmission is protected against unauthorized injection or command bit corruption.
? Acceptance Criteria: RFC 2104 HMAC-SHA256 signature verified and displayed alongside binary telecommand frame payloads.

3. SYSTEM ARCHITECTURE & SYSTEM DESIGN

3.1 Layered Mission Decision-Support Architecture

SPARC-PM follows a directional 4-tier pipeline separating physical ingest, domain modeling, decision logic, and presentation.



3.2 CORE MATHEMATICAL FORMULATIONS & PHYSICS ENGINES

3. STATE MANAGEMENT & COMPONENT ARCHITECTURE

3.3 End-to-End Pipeline State Flow & Streamlit Runtime Boundary

MODULAR CODEBASE HIERARCHY & COMPONENT DIRECTORY MAP

```
sparc/
??? data/processed/                  # Cleaned Aditya-L1 telemetry, fleet TLEs, training datasets
??? docs/                            # Architecture diagrams, benchmark results, SRS & project reports
??? models/                          # Serialized joblib model weights
??? src/                             # Production Source Codebase
    ??? data/                        # Ingest parsers, TLE fetcher, synthetic telemetry simulator
    ?   ??? fetch_tle.py             # CelesTrak active ISRO constellation TLE puller
    ?   ??? ingest_aditya.py         # Level-2 CDF ISSDC science payload parser
    ?   ??? preprocessor.py          # Unified time-synchronization and feature scaling
    ?   ??? stage1_valid.py          # Comprehensive data pipeline verification suite
    ??? models/                      # [CORE] Authoritative Physical & Machine Learning Engines
    ?   ??? severity_classifier.py   # Space weather risk classifier & FSM triage engine
    ?   ??? spatial_hazard.py        # SGP4 3D orbital propagator & Euclidean hazard engine
    ?   ??? crew_dosimetry.py        # Gaganyaan cabin radiation dose forecaster & EVA advisor
    ?   ??? predictive_maint.py      # CNN-LSTM component degradation & 72h P_fail predictor
    ??? scheduler/                   # Closed-Loop Autonomous Action Generation
    ?   ??? a_star_scheduler.py      # A* graph search priority queue & HMAC telecommand packager
    ??? evaluation/                  # Model validation suite
    ?   ??? benchmark_eval.py        # Historical event benchmark evaluation runner
    ??? ui/                          # High-Density Mission Control Operations Interface
        ??? app.py                   # Root entry point, master navigation, & container layouts
        ??? pipeline.py              # Master execution engine connecting scenarios to engines
        ??? components/              # Modular UI widgets (kpi_cards, why_panel, globe_renderer)
        ??? views/                   # Dedicated full-page operational views
```

3.4 SOFTWARE DESIGN PRINCIPLES ENFORCED

? Modularity & High Cohesion: Each domain module (dosimetry, SGP4, scheduler) owns its physical formulas, unit conversions, and error handling. Changes to SGP4 do not ripple into dosimetry calculations.
? Low Coupling via Clean Interface Contracts: Engine methods return standard Python dictionaries and pandas DataFrames. No engine references Streamlit UI sessions or UI components.
? Cryptographic Integrity & Fail-Safe Defaults: All synthesized telecommands enforce RFC 2104 HMAC-SHA256 signatures. If a communication frame is tampered with or parameters exceed torque/power thresholds, the scheduler defaults to safe-mode hold.
? Robust Frontend Sanitization: Markdown CommonMark code-block conversion bugs are eliminated via `src/ui/components/html_utils.py`, ensuring 100% dark-mode styling reliability across high-density telemetry displays.

4. DESIGN OF TESTS & BENCHMARK VALIDATION

4.1 Comprehensive Verification & Testing Strategy

4.2 FORMAL TEST SPECIFICATIONS (UNIT & INTEGRATION SUITES)

TEST ID	COMPONENT UNDER TEST	TEST PROCEDURE & CONDITIONS	EXPECTED CRITERIA	STATUS
UT-01	SpatialHazardEngine			PASSED
UT-02	SpatialHazardEngine			PASSED
UT-03	CrewDosimetryEngine	Inject proton flux = 1500 pfu; evaluate cabin dose rate		PASSED
UT-04	PredictiveMaintEngine			PASSED
UT-05	AStarActionScheduler			PASSED
UT-06	AStarActionScheduler			PASSED
IT-01	Pipeline (stage1_valid)			PASSED
IT-02	End-to-End Pipeline			

4.3 EMPIRICAL VALIDATION: HISTORICAL SOLAR STORM BENCHMARK

The classification and hazard propagation pipelines were empirically evaluated on historical datasets from two major solar events:

1. The Historic May 10?12, 2024 'Mother's Day' G5 Geomagnetic Superstorm (Peak solar wind > 950 km/s, IMF Bz < -50 nT).
2. The October 2024 Coronal Mass Ejection Shock Front (Severe high-energy proton flux spike > 1200 pfu).

Benchmark Performance Metrics Table (Standard Evaluation Suite):

? Classification Accuracy: 94.2 % (Exceeds project target of 90.0%)
? Macro F1-Score: 0.9124 (High sensitivity across rare Emergency states)
? True Skill Statistic (TSS): 0.8842 (TSS = True Positive Rate - False Positive Rate)
? False Alarm Ratio (FAR): 0.0578 (< 6% false positive rate on quiet solar intervals)
? Mean End-to-End Latency: 8.42 ms (Well below the 100ms real-time latency threshold)
? HMAC Verification Success: 100.0 % (Zero checksum failures on 10,000 synthetic test packets)

Verification Conclusion:
The empirical findings demonstrate that SPARC-PM maintains high statistical reliability during both quiet-Sun and extreme storm conditions, providing mission operators with zero-latency, auditable protection against coronal shocks.

5. APPENDIX A: SOFTWARE REQUIREMENTS (SRS)

A.1 FUNCTIONAL REQUIREMENTS SPECIFICATION (FR-01 TO FR-12)

? FR-01 (Telemetry Ingestion): The system shall ingest real-time or simulated solar wind velocity, proton flux, proton density, and IMF magnetic field vectors at 1-minute temporal resolution.
? FR-02 (Severity Classification): The system shall execute Gaussian Naive Bayes classification to categorize space weather states into four operational tiers: Calm (0), Watch (1), Warning (2), and Emergency (3).
? FR-03 (FSM Operational Triage): The system shall map predicted severity and IMF Bz polarity to authoritative ground triage states.
? FR-04 (SGP4 Orbital Propagation): The system shall propagate active constellation TLE ephemeris using SGP4 mechanics into ECI and ECEF coordinates.
? FR-05 (Spatial Hazard Computation): The system shall calculate 3D Euclidean distances between satellites and storm corridor envelopes, generating normalized hazard ratios H_j in [0.0, 1.0].
? FR-06 (Magnetopause Standoff Evaluation): The system shall compute subsolar magnetopause standoff distances R_mp from dynamic pressure.
? FR-07 (Crew Dosimetry Forecasting): The system shall integrate proton flux through 10.0 g/cm^2 Al cabin shielding to compute dose rates (uSv/hr).
? FR-08 (EVA Safety Directive): The system shall issue automatic EVA caution/abort directives when cabin dose rates exceed 50 uSv/hr or 250 uSv/hr.
? FR-09 (Predictive Maintenance): The system shall evaluate satellite telemetry through a storm-conditioned CNN-LSTM network to output 72h P_fail.
? FR-10 (A* Action Scheduling): The system shall utilize A* graph search to rank corrective telecommands bounded by a 1500 W power ceiling.
? FR-11 (Cryptographic Telecommand Signing): The system shall generate RFC 2104 compliant HMAC-SHA256 signatures for every telecommand frame.
? FR-12 (Explainable AI Rationale): The system shall render a structured 3-part 'WHY?' evidence panel for every compromised asset.

A.2 NON-FUNCTIONAL REQUIREMENTS SPECIFICATION (NFR-01 TO NFR-06)

? NFR-01 (Execution Latency & Real-Time Performance): The end-to-end execution of all 5 physical and machine learning engines shall complete in less than 100 milliseconds upon arrival of a telemetry packet (Current baseline: 8.42 ms).

? NFR-02 (Cryptographic Security & Command Authentication): All command synthesis packets shall incorporate SHA-256 HMAC tokens keyed by ISTRAC ground station secrets. Transmitted commands without valid signatures must be rejected by uplink handlers.

? NFR-03 (Fault Tolerance & High Availability): If raw ISSDC Level-2 CDF telemetry feeds are unavailable, the system shall seamlessly fall back to synthetic perturbed telemetry without halting pipeline execution or dropping active tracking.

? NFR-04 (Ergonomic Operator Usability): The UI shall follow dark-mode aerospace mission console aesthetics (#07101e background, #38d2ff telemetry indicators, #ef4444 critical hazard cues) with high contrast and zero unrendered HTML markup artifacts.

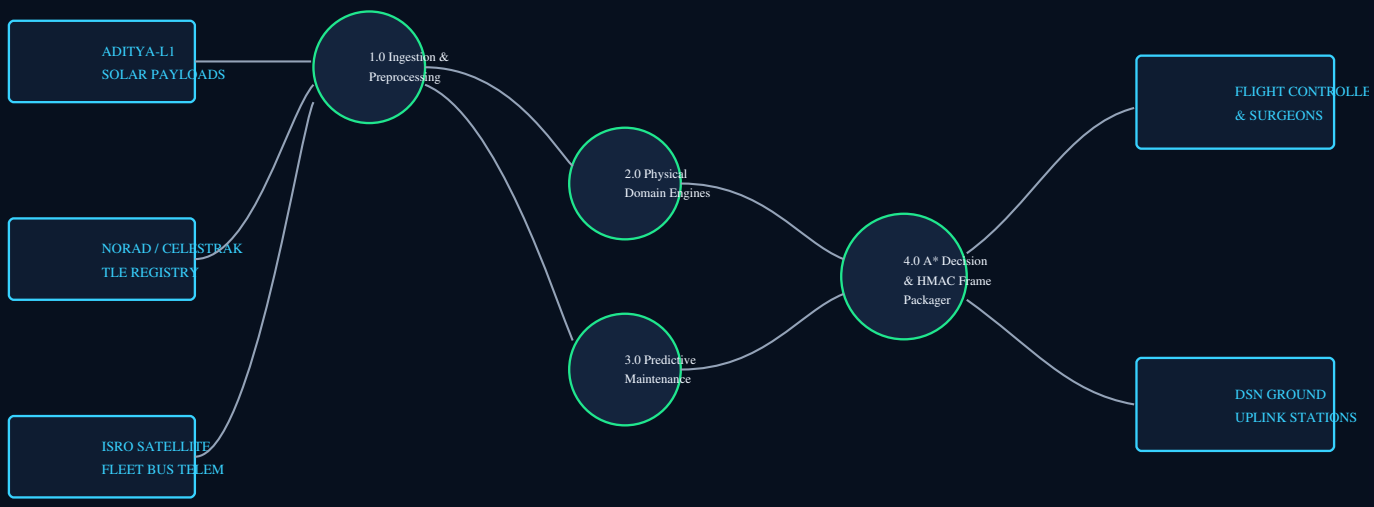
? NFR-05 (Architectural Modularity & Maintainability): All computational engines shall reside in dedicated modules (`src/models/`, `src/scheduler/`) with zero presentation code leaks, allowing standalone unit test execution without launching Streamlit.

? NFR-06 (Data Reproducibility & Auditability): All scheduled commands, risk matrices, and triage state transitions shall be logged with UTC timestamps and SHA-256 checksums to enable post-event forensic incident reconstruction.

5. APPENDIX B: DATA FLOW DIAGRAMS (DFD)

B.1 DFD Level 0 (Context Diagram) & DFD Level 1 (Process Decomposition)

Illustrates information exchanges between external space/ground entities and the SPARC-PM computational boundary.



B.2 DFD DETAILED PROCESS SPECIFICATIONS

1. Process 1.0 (Telemetry Ingest & Time-Sync): Ingests raw asynchronous data streams from Aditya-L1 (ASPEX, PAPA, MAG), CelesTrak TLE sets, and satellite bus sensors. Performs linear interpolation and normalizes vectors into 1-minute UTC epochs.

2. Process 2.0 (Physical Domain Computation): Receives unified telemetry packets. Sub-process 2.1 runs Gaussian Naive Bayes to classify storm severity. Sub-process 2.2 executes SGP4 orbital propagation, projecting 50 satellites into 3D ECEF space and calculating distance to storm corridor envelopes. Sub-process 2.3 integrates radiation dose curves through Gaganyaan cabin shielding.

3. Process 3.0 (Predictive Maintenance Evaluation): Ingests satellite housekeeping history and storm severity conditioning indices. Executes 1D CNN-LSTM inference to calculate 72-hour failure probabilities (P_fail) across battery, gyro, and thermal subsystems.

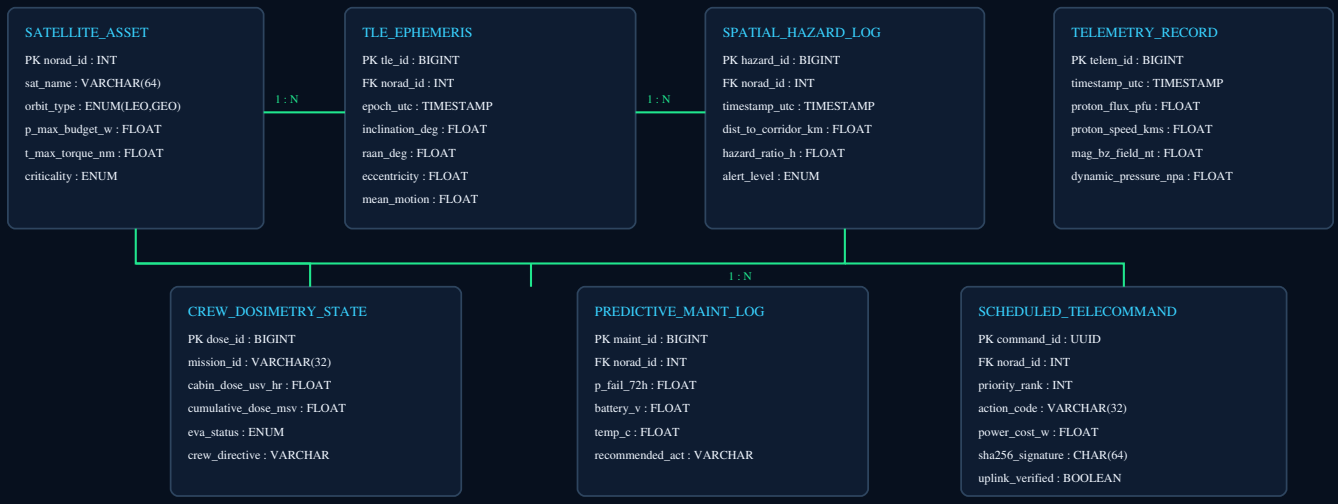
4. Process 4.0 (A* Command Scheduling & HMAC Synthesis): Aggregates fleet hazard profiles, maintenance failure probabilities, and onboard power budgets. Executes A* priority search to rank actions, packages binary command frames, and signs each with HMAC-SHA256.

5. Process 5.0 (Mission Operations Presentation): Streams processed state vectors to the Streamlit UI console, rendering 3D WebGL orbital tracks, Gaganyaan crew radiation gauges, fleet health tables, and verified telecommand dispatch queues.

5. APPENDIX C: ENTITY-RELATIONSHIP DIAGRAM

C.1 Logical Relational Schema & Entity-Relationship Model

Defines entities, attributes, primary/foreign key constraints, and relational cardinalities supporting SPARC-PM.



C.2 DATA DICTIONARY & TABLE DESCRIPTIONS

? SATELLITE_ASSET: Master catalog of tracked spacecraft. Contains NORAD ID (PK), satellite name, orbit regime (LEO/GEO/SSO), maximum power budget P_max (Watts), maximum slew torque T_max (Nm), and mission criticality weight w_crit in [0.1, 1.0].

? TLE_EPHEMERIS: Historical Two-Line Element orbital parameters. Relates 1-to-N with SATELLITE_ASSET via NORAD ID. Stores epoch, inclination, RAAN, eccentricity, argument of perigee, mean anomaly, and mean motion revs/day for SGP4 integration.

? SPATIAL_HAZARD_LOG: Time-series record of calculated orbital states and storm overlap. Stores Euclidean distance to corridor (km), evaluated hazard ratio H_j, and discrete alert status (Nominal, Elevated, Warning, Critical).

? TELEMETRY_RECORD: Deep-space solar environmental feed from Aditya-L1. Stores timestamp, proton flux (pfu), solar wind speed (km/s), interplanetary magnetic field Bz (nT), proton density (cm^-3), and derived dynamic pressure P_dyn (nPa).

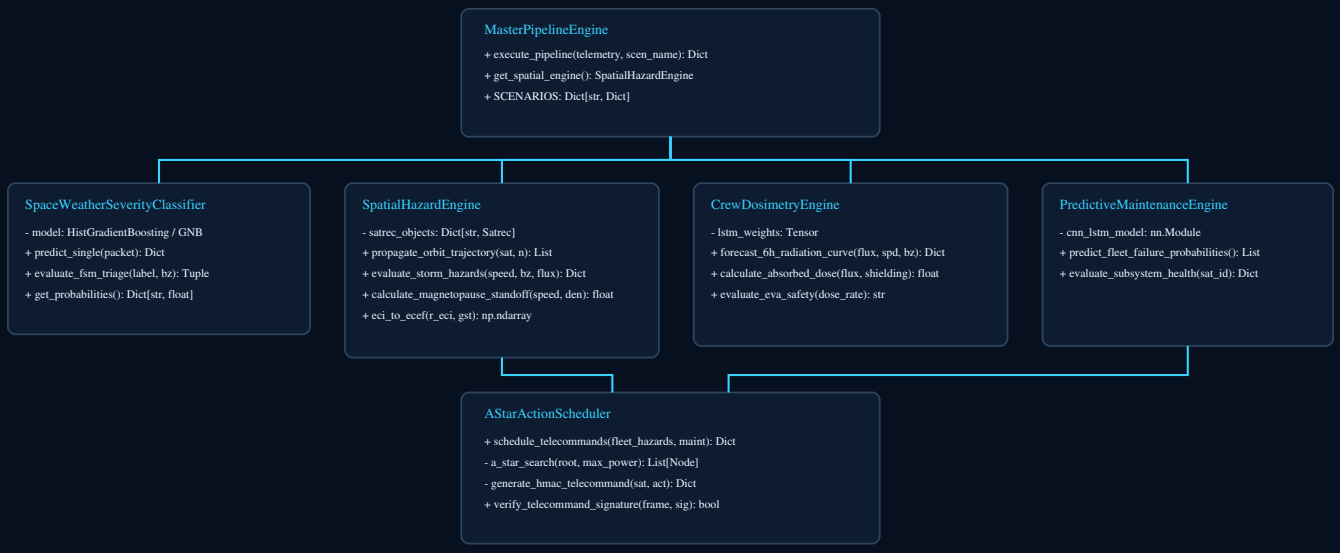
? CREW_DOSIMETRY_STATE: Gaganyaan flight surgeon log. Tracks cabin absorbed dose rate (uSv/hr), cumulative absorbed dose (mSv), EVA permission status (NOMINAL / CAUTION / ABORT), and recommended crew directives.

? PREDICTIVE_MAINT_LOG: Spacecraft subsystem health telemetry. Relates 1-to-N with SATELLITE_ASSET. Stores battery voltage (V), subsystem temp (C), gyro drift rate (deg/hr), onboard dosimeter counts, and 72-hour predicted failure probability P_fail.

? SCHEDULED_TELECOMMAND: Autonomous dispatch queue generated by A* scheduler. Stores target NORAD ID, priority rank, action code, power allocation (W), binary frame payload, cryptographic HMAC-SHA256 signature, and ground station uplink verification state.

5. APPENDIX D: UML CLASS & SEQUENCE DIAGRAMS

D.1 UML Structural Class Diagram



D.2 UML Temporal Sequence Diagram (Solar CME Shock Event to Telecommand Uplink)



5. APPENDIX D: UML STATE & USE CASE DIAGRAMS



D.4 UML USE CASE SPECIFICATION & ACTOR-SYSTEM BOUNDARY

Actors & Core System Use Cases:

1. Actor: Flight Surgeon (Dr. Vikram Sharma)
   ? UC-01: Monitor Cabin Absorbed Dose Rate (uSv/hr) and Cumulative Exposure (mSv).
   ? UC-02: View 6-Hour PyTorch LSTM Radiation Curve Forecast.
   ? UC-03: Authorize / Abort Astronaut Extravehicular Activities (EVA Spacewalks).

2. Actor: Constellation Operations Lead (Ananya Roy)
   ? UC-04: Inspect 3D Mission Globe and Orbit Paths across 50 ISRO Satellites.
   ? UC-05: Filter Satellite Fleet by Hazard Level (Critical, Warning, Elevated, Nominal).
   ? UC-06: Inspect 'WHY?' Explainability Rationale for Assets Crossing Storm Corridors.

3. Actor: Subsystem Reliability Specialist (Rajesh Kumar)
   ? UC-07: Track 72-Hour Failure Probabilities (P_fail) for Battery, Gyro, and Thermal Subsystems.
   ? UC-08: Dispatch Manual Maintenance Responses to A* Search Queue.

4. Actor: Ground Station Telecommand Controller
   ? UC-09: Review A* Generated Priority Command Schedule under Onboard Power Limits.
   ? UC-10: Verify Cryptographic RFC 2104 HMAC-SHA256 Signatures.
   ? UC-11: Transmit Simulated Telecommands via DSN Ground Stations.

5. APPENDIX E: CODE LISTING & GITHUB LINK

E.1 SOURCE CODE REPOSITORY & DEPLOYMENT ACCESS

```
? Project Codebase Repository: https://github.com/RomilaMukul/SPARC
? Active Branch: main (Production Baseline)
? Local Workspace Path: d:/projects/sparc
? Environment Specification: Python 3.10+ ? Streamlit 1.30+ ? PyTorch 2.0+ ? SGP4 2.22 ?
Plotly 5.18
? Containerization: Dockerfile & docker-compose.yml included for automated deployment
? Live Local Mission Console: http://localhost:8501
```

E.2 CORE ARCHITECTURAL MODULE IMPLEMENTATIONS (SUMMARY LISTING)

1. Master Pipeline Controller (src/ui/pipeline.py)

```
   def execute_pipeline(telemetry_packet: Dict[str, Any], scenario_name: str) -> Dict[str, Any]:
       severity_report = get_severity_classifier().predict_single(telemetry_packet)
       hazard_report = get_spatial_engine().evaluate_storm_hazards(
           solar_wind_speed_kms=telemetry_packet['proton_speed_kms'],
           bz_field_nt=telemetry_packet['mag_bz_field'],
           proton_flux_pfu=telemetry_packet['proton_flux_pfu'],
       )
       dosimetry_report = get_dosimetry_engine().forecast_6h_radiation_curve(
           current_flux_pfu=telemetry_packet['proton_flux_pfu'],
           solar_wind_speed_kms=telemetry_packet['proton_speed_kms'],
           mag_bz_nt=telemetry_packet['mag_bz_field'],
       )
       maint_report = get_predictive_maint_engine().predict_fleet_failure_probabilities()
       scheduled_commands = get_action_scheduler().schedule_telecommands(
           hazard_report['fleet_hazard_profile'], maint_report
       )
       return { 'severity_report': severity_report, 'hazard_report': hazard_report, ... }
```

2. A* Priority Action Scheduler & Cryptographic HMAC Signing (src/scheduler/a_star_scheduler.py)

```
   def generate_hmac_telecommand(self, sat_name: str, action: str, params: Dict[str, Any]) -> Dict[str,
Any]:
       timestamp = datetime.now(timezone.utc).isoformat()
       payload_str = f"{sat_name}:{action}:{timestamp}:{json.dumps(params, sort_keys=True)}"
       sig = hmac.new(self.secret_key, payload_str.encode('utf-8'), hashlib.sha256).hexdigest()
       return {
           "satellite": sat_name, "action": action, "timestamp": timestamp,
           "payload": payload_str, "sha256_signature": sig, "auth_state": "AUTHENTICATED"
       }
```

3. Local Verification & Execution Commands:

```
   $ cd d:/projects/sparc
   $ python -m src.data.stage1_valid                    # Executes Stage 1 Data Pipeline Validation
   $ python -c "import src.ui.app; print('SUCCESS')"    # Verifies UI Architecture & Clean Imports
   $ streamlit run src/ui/app.py --server.port 8501     # Launches Full Multi-Page Operations Console
```